

Cedrick TCHAKONTE TCHIEGUE

Student at ENSTA, having completed the 2nd year of the engineering cycle (specialization in Artificial Intelligence and Cyber-Physical Systems), currently on a gap year. Looking for a 6-month internship to strengthen my background in AI and machine learning, discover new fields of innovation, and contribute to meaningful projects.

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Education

ENSTA Campus de Paris-Saclay, Palaiseau, France Aug 2024 – present

- Second-year engineering student in Artificial Intelligence and Cyber-Physical Systems
- **Coursework** : Machine learning, Image Recognition and analysis, microprocessor architecture, parallel and distributed systems, statistical learning, control theory, robotic navigation.

National Advanced School of Engineering, Yaoundé, Cameroon Sept 2020 – Jul 2024

- Intensive preparatory program in mathematics and physical sciences for competitive engineering entrance examinations
- **Coursework** : Linear algebra, electromagnetism, mechanics, differential equations, probability and statistics, thermodynamics.

Experience

Junior Research Engineer Intern, Objectware, Paris, France Sept 2025 – present

- Conducted R&D on an AI-based decision support system for enterprise applications, designing ML models for predictive analytics to improve decision accuracy and efficiency.
- Developed AI algorithms for business intelligence, supporting enterprise clients with advanced analytics and decision-making tools.
- Worked with cross-functional teams (market directors, HR, consultants, CEO) to integrate AI solutions, improving efficiency and streamlining workflows.

Research and Development Intern, STMicroelectronics, Grenoble, France May 2025 – Aug 2025

- Worked within the System Level Modeling team (MDRF division) on C++-based "Digital Twin" simulations enabling virtual execution of embedded software for multiple ST divisions (automotive, RF, secure MCUs, etc.).
- Developed AI-assisted workflows for documentation analysis and model generation, using LLMs, RAG, and vector databases to process complex simulation codebases.
- Benchmarked and integrated the solution into SOC Digital Twin development, improving documentation retrieval efficiency and enhancing productivity in simulation-driven validation.

Projects

Retinal Vessel Segmentation in SLO Ophthalmoscopy, ENSTA Campus de Paris-Saclay May 2025 – Jun 2025

- Developed automated retinal vessel segmentation system (Computer Vision, Biomedical Imaging) using image derivation operators and specialized vascular filters for Scanning Laser Ophthalmoscopy (SLO) retinal images, implementing two segmentation approaches against IOSTAR database ground truth.
- Applied advanced image processing techniques including gradient operators and morphological filtering for precise vascular structure detection and effective vessel network extraction.

Computer Vision and Feature Detection Projects, ENSTA Campus de Paris-Saclay Mar 2025 – May 2025

- Image Segmentation : Implemented Bayesian classification and K-means clustering for skin detection using Essex dataset, achieving robust pixel-level classification.
- Feature Detection : Developed feature detection pipeline using Harris corner detection, ORB, and KAZE algorithms with OpenCV, applying gradient analysis and morphological operations for robust point matching across scales and transformations.

4D GPS Navigation System for VTOL Aircraft, ENSTA Campus de Paris-Saclay Oct 2024 – May 2025

- Developed trajectory optimization algorithms for vertical takeoff and landing aircraft in a project with the company Technoplane, using 3D mapping and dynamic weather modeling as a 4th dimension. Simulated and analyzed real-time path planning strategies.

Certifications

Deep Learning Specialization, DeepLearning.AI

Improving Deep Neural Networks, Structuring ML Projects, Convolutional Neural Networks, Sequence Models.

Project Management MOOC Session 24 – Classic Track

Online course on project management methodologies, covering strategic planning, execution, monitoring, and risk management.

Languages and Skills

- **Languages** : French (Native), English (Fluent - B2)
- **Programming & Software Engineering** : Python, C/C++, Java, JavaScript, R, Bash, MATLAB
- **Artificial Intelligence & Machine Learning** : TensorFlow, PyTorch, Keras, Scikit-Learn, OpenCV, Hugging Face, Transformers, LangChain

Volunteer Work, Extracurricular Activities and Personal interests

Finance Officer, ENSTAFRIK Association, ENSTA Apr 2025 – present

Management of financial operations, preparation of budgets and coordination of cultural events.

Volunteer, Ryz'Ôm Joëlettes (Disability Inclusion Association) Nov 2024 – present

Actively contributing to community support and inclusion initiatives with 30+ volunteers, assisting up to 10 individuals with mobility disabilities during running events using joëlettes, and leading high school training sessions where I teach students to assemble and operate joëlettes while promoting inclusion.

Personal interests : volleyball, tennis, chess.